

STA623 - Bayesian Data Analysis

Practical 3: Monte Carlo, and writing your own MCMC samplers

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Goals. Quantify Monte Carlo error; use simulation for posterior predictive checks; implement a Metropolis sampler for a two-parameter model and a Gibbs sampler for the normal model; assess your own samplers with multiple chains. Accompanies Lecture 3.

Exercise 1: Monte Carlo error

The posterior for the malaria-episode rate in Lecture 2 was $\lambda \mid \mathbf{y} \sim \text{Gamma}(45, 20)$.

- (a) For $S = 10, 100, 1000, 10^4, 10^5$, estimate $E[\lambda \mid \mathbf{y}]$ and $P(\lambda < 2.1 \mid \mathbf{y})$ by Monte Carlo, together with their Monte Carlo standard errors (MCSE). Compare with the exact values.
- (b) How many draws are needed for the MCSE of $P(\lambda < 2.1 \mid \mathbf{y})$ to be below 0.005 (so that the second decimal place is reliable)? Derive it from the formula, then check by simulation.
- (c) Repeat the $S = 1000$ estimate of the mean 500 times. Compare the standard deviation of the 500 estimates with the MCSE formula.

Exercise 2: posterior predictive simulation and a first model check

Continue with the GSS data of Practical 2 (women without a degree, posterior $\theta_1 \sim \text{Gamma}(219, 112)$). The observed distribution of the number of children in this group was:

```
obs_counts <- tibble(children = 0:6, mothers = c(20, 19, 38, 20, 10, 2, 2))
kable(obs_counts)
```

children	mothers
0	20
1	19
2	38
3	20
4	10
5	2

children	mothers
6	2

```
y_obs <- rep(obs_counts$children, obs_counts$mothers)
```

- Simulate the posterior predictive distribution for the number of children of one new woman and plot it next to the observed relative frequencies.
- Let $t(\mathbf{y})$ be the number of women with exactly 2 children divided by the number with exactly 1 child; here $t(\mathbf{y}) = 38/19 = 2$. Simulate 10,000 **replicated datasets** of 111 women, each using a *new* posterior draw of θ_1 , and compute t for each. Estimate $P(t(\mathbf{y}^{\text{rep}}) \geq 2 \mid \mathbf{y})$.
- What does this say about the Poisson model?

Exercise 3: a Metropolis sampler for Poisson regression

Malaria episodes for 60 children, with an indicator of whether the household has an insecticide-treated net:

```
set.seed(31)
itn <- rep(0:1, each = 30)
episodes <- rpois(60, exp(0.8 - 0.6 * itn))
tapply(episodes, itn, mean)
```

```
      0      1
2.000000 1.233333
```

Model: $Y_i \sim \text{Pois}(\mu_i)$, $\log \mu_i = \alpha_0 + \alpha_1 \text{itn}_i$, priors $\alpha_0 \sim \mathcal{N}(0, 1.5^2)$, $\alpha_1 \sim \mathcal{N}(0, 1)$.

- Write a function `log_post(alpha)` returning the log of likelihood $\times$ prior for $\alpha = (\alpha_0, \alpha_1)$.
- Write a random-walk Metropolis sampler that proposes $\alpha^* \sim \mathcal{N}_2(\alpha^{(s-1)}, \delta^2 \mathbf{I})$. Run 4 chains of 5,000 iterations from dispersed starting values with $\delta = 0.15$. Report the acceptance rate.
- Plot trace plots of all chains. Discard a burn-in of 1,000 iterations, then compute $\widehat{R}$ and bulk ESS with the `posterior` package.
- Report the posterior median and 95% credible interval of the rate ratio e^{α_1} and $P(e^{\alpha_1} < 1 \mid \mathbf{y})$. Compare with `glm(episodes ~ itn, family = poisson)`.
- Try $\delta = 0.01$ and $\delta = 1$. What happens to the acceptance rate and ESS?

Exercise 4: a Gibbs sampler for the normal model

Haemoglobin (g/dL) for 25 pregnant women:

```
hb <- c(10.2, 11.8, 9.4, 12.1, 10.9, 11.3, 8.7, 10.5, 12.6, 11.0, 9.9, 10.8, 11.5,  
        10.1, 9.2, 12.0, 10.7, 11.2, 9.8, 10.4, 13.1, 10.0, 11.6, 9.5, 10.9)
```

Model: $Y_i \sim \mathcal{N}(\mu, \sigma^2)$ with independent priors $\mu \sim \mathcal{N}(12, 2^2)$, $1/\sigma^2 \sim \text{Gamma}(\nu_0/2, \nu_0\sigma_0^2/2)$ with $\nu_0 = 1$, $\sigma_0^2 = 2$.

- (a) Write down the two full conditional distributions (see Lecture 2) and implement a Gibbs sampler.
- (b) Run 4 chains of 3,000 iterations from dispersed starting values; check convergence with trace plots and $\widehat{R}$.
- (c) Anaemia in pregnancy is defined as Hb < 11 g/dL. Estimate $P(\mu < 11 \mid \mathbf{y})$, and the posterior predictive probability that a *new* woman from this population is anaemic. Why do these differ?
- (d) Estimate the posterior median and 95% interval of the coefficient of variation σ/μ .